Moment	Correlation with top-random interpretability score
Mean	-0.09
Variance	0.02
Skew	0.20
Kurtosis	0.15

Table 2: Correlation of interpretability score with feature moments across residual stream results, all layers, with dictionary size ratios $R \in \{0.5, 1, 2, 4, 8\}$.

This also explains why Independent Component Analysis (ICA), which maximises the non-Gaussianity of the found components, is the best performing of the alternatives that we considered.

D Qualitative Feature Analysis

D.1 Residual Stream Basis

Figure 11 gives a token activation histogram of the residual stream basis. Connecting this residual stream dimension to the apostrophe feature from Figure 4, this residual dimension was the 10th highest dimension read from the residual stream by our feature[7].

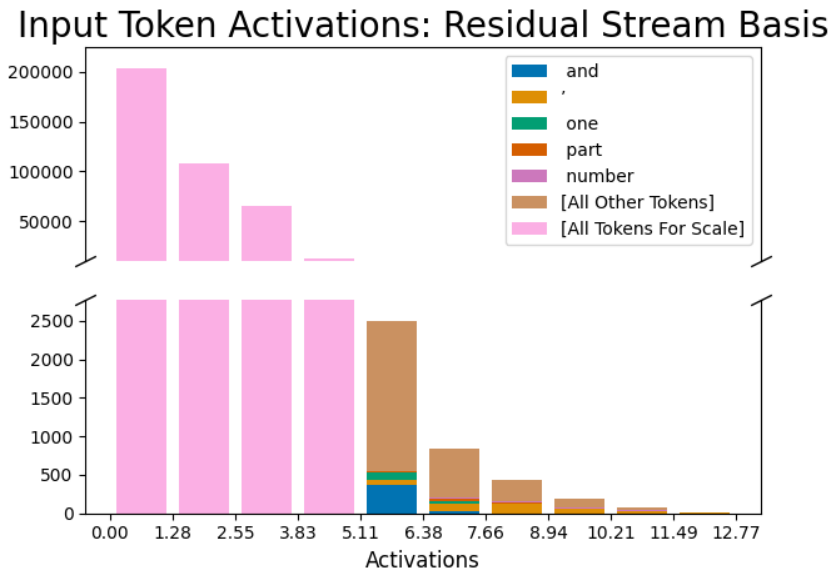

Figure 11: Histogram of token counts in the neuron basis. Although there are a large fraction of apostrophes in the upper activation range, this only explains a very small fraction of the variance for middle-to-lower activation ranges.

D.2 Examples of Learned Features

Other features are shown in Figures 12, 13, 14, and 15.

D.3 Feature Search Details

We searched for the apostrophe feature using the sentence " I don't know about that. It is now up to Dave"', and seeing which feature (or residual stream dimension) activates the most for the last apostrophe token. The top activating feature in our dictionary was an outlier dimension feature (i.e., a feature direction that mainly reads from an outlier dimension of the residual stream), the apostrophes

[7]The first 9 did not have apostrophes in their top-activations like dimension 21.

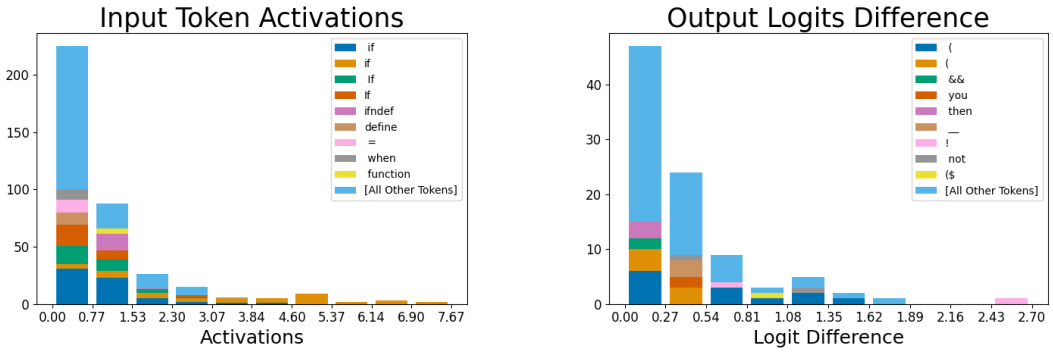

Figure 12: ‘If’ feature in coding contexts

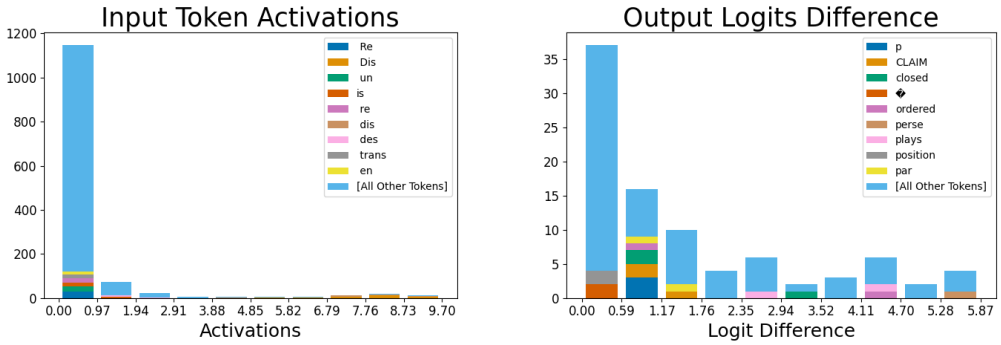

Figure 13: ‘Dis’ token-level feature showing bigrams, such as ‘disCLAIM’, ‘disclosed’, ‘disordered’, etc.

after O (and predicted O’Brien, O’Donnell, O’Connor, O’clock, etc), then the apostrophe-preceding-s feature.

For the residual basis dimension, we searched for max and min activating dimensions (since the residual stream can be both positive and negative), where the top two most positive dimensions were outlier dimensions, the top two negative dimensions were our displayed one and another outlier dimension, respectively.

D.4 Failed Interpretability Methods

We attempted a weight-based method going from the dictionary in layer 4 to the dictionary in layer 5 by multiplying a feature by the MLP and checking the cosine similarity with features in layer 5. There were no meaningful connections. Additionally, it’s unclear how to apply this to the Attention sublayer since we’d need to see which position dimension the feature is in. We expected this failed by going out of distribution.

E Number of Active Features

In Figure 16 we see that, for residual streams, we consistently learn dictionaries that are at least 4x overcomplete before some features start to drop out completely, with the correct hyperparameters. For MLP layers you see large numbers of dead features even with hyperparameter $\alpha = 0$. These figures informed the selection of $\alpha = 8.6e-4$ and $\alpha = 3.2e-4$ that went into the graphs in Section 3 for the residual stream and MLP respectively. Due to the large part of the input space that is never used due to the non-linearity, it is much easier for MLP dictionary features to become stuck at a position where they hardly ever activate. In future we plan to reinitialise such ‘dead features’ to ensure that we learn as many useful dictionary features as possible.

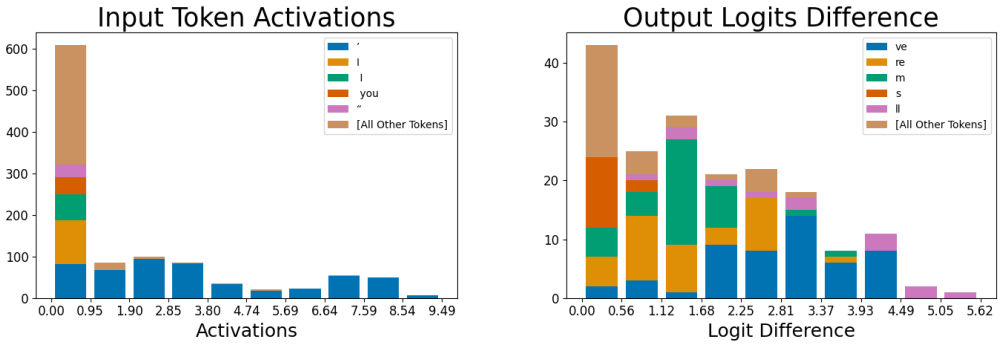

Figure 14: Apostrophe feature in “I’ll”-like contexts.

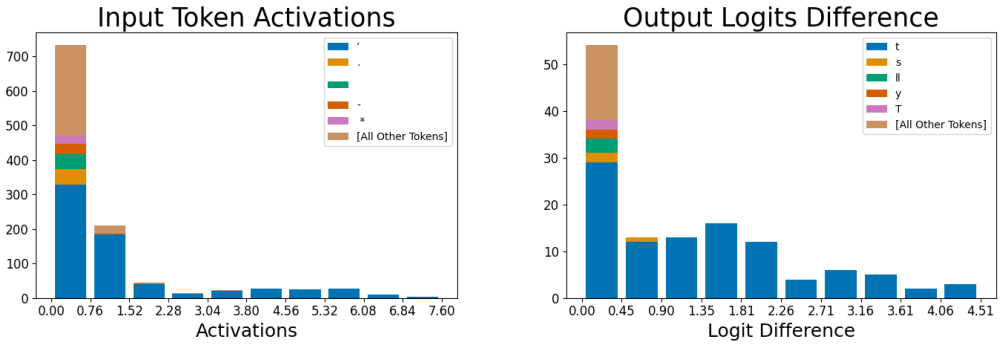

Figure 15: Apostrophe feature in “don’t”-like contexts.

F Editing IOI Behaviour on other Layers

In Figure 17 we show results of the procedure in Section 4 across a range of layers in Pythia-410M.

G Top K Comparisons

As mentioned in Section 3, the comparison directions learnt by sparse coding and those in the baselines are not perfectly even. This is because, for example, a PCA direction is active to an entire half-space on one side of a hyperplane through the origin, whereas a sparse coding feature activates on less than a full direction, being only on the far side of a hyperplane that does not intersect the origin. This is due to the bias applied before the activation, which is, in practice, always negative. To test whether this difference is responsible for the higher scores, we run a variant of PCA and ICA in which we have a fixed number of directions, K, which can be active for any single datapoint. We set this K to be equal to the average number of active features for a sparse coding dictionary with ratio $R = 1$ and $\alpha = 8.6e - 4$ trained on the layer in question. We compare the results in Figure 18, showing that this change does not explain more than a small fraction of the improvement in scores.